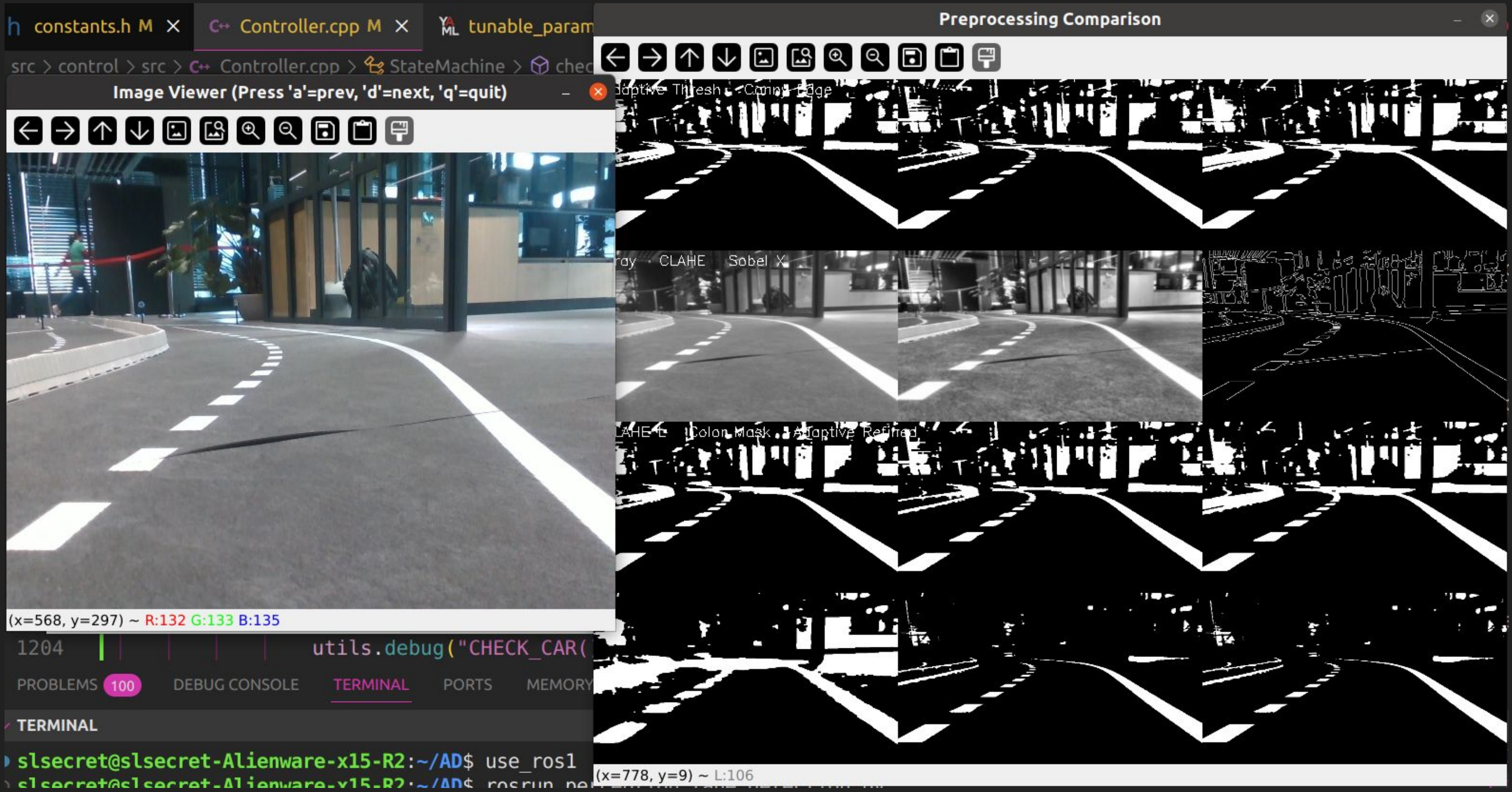
Software PDR March 23

simon

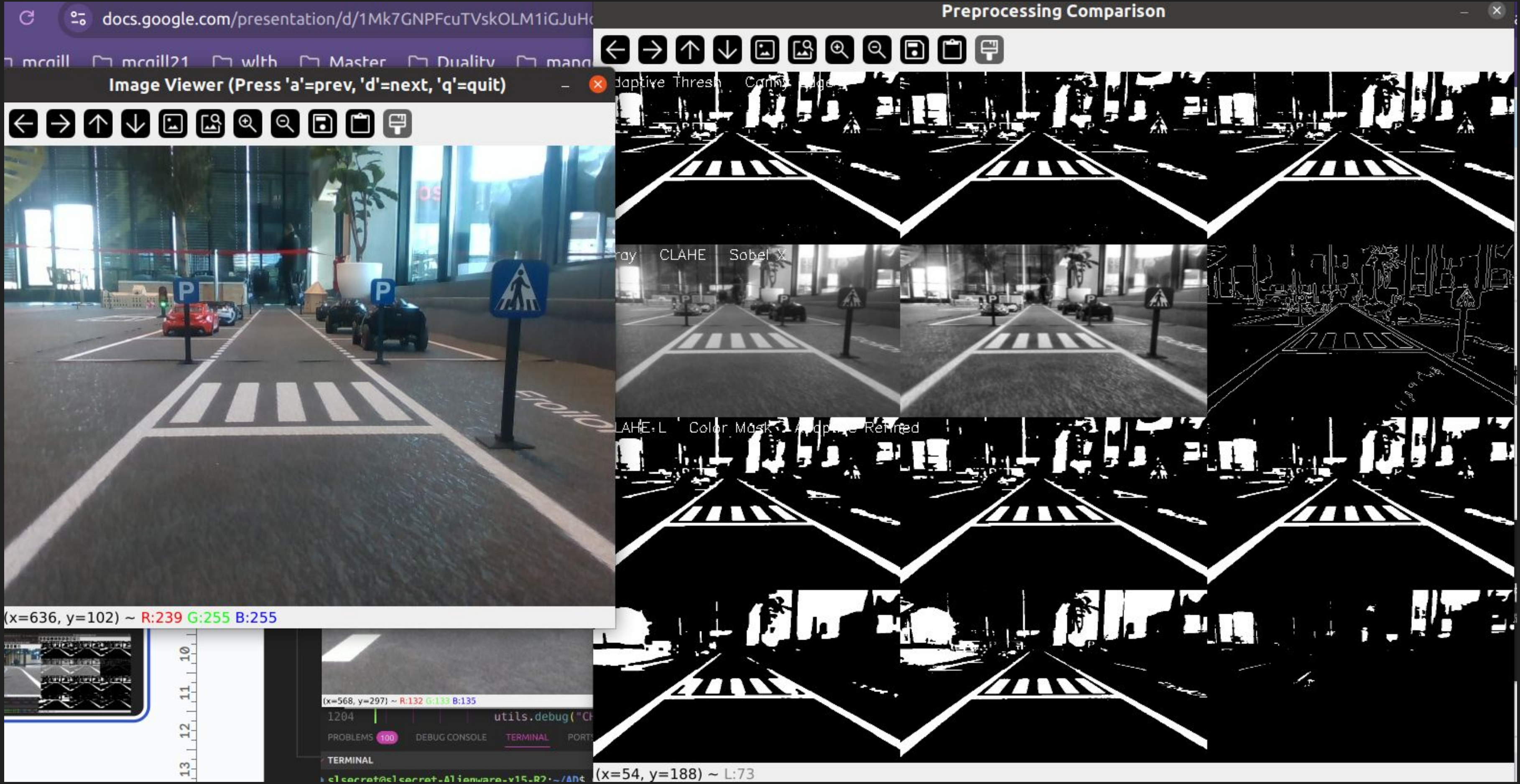
Lane

- adaptive threshold
- pixel to meter
- simulator testing
-

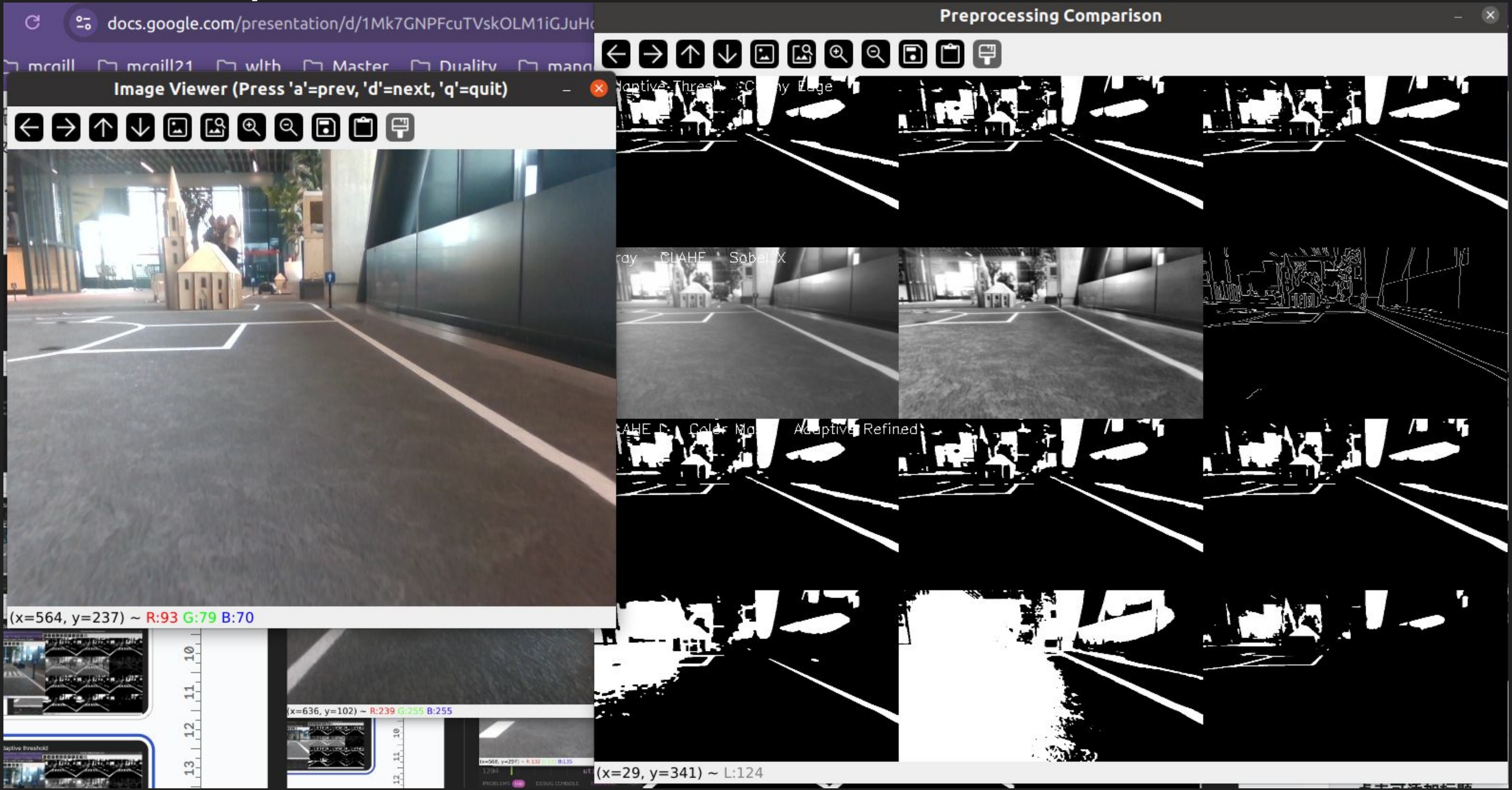
Lane: adaptive threshold



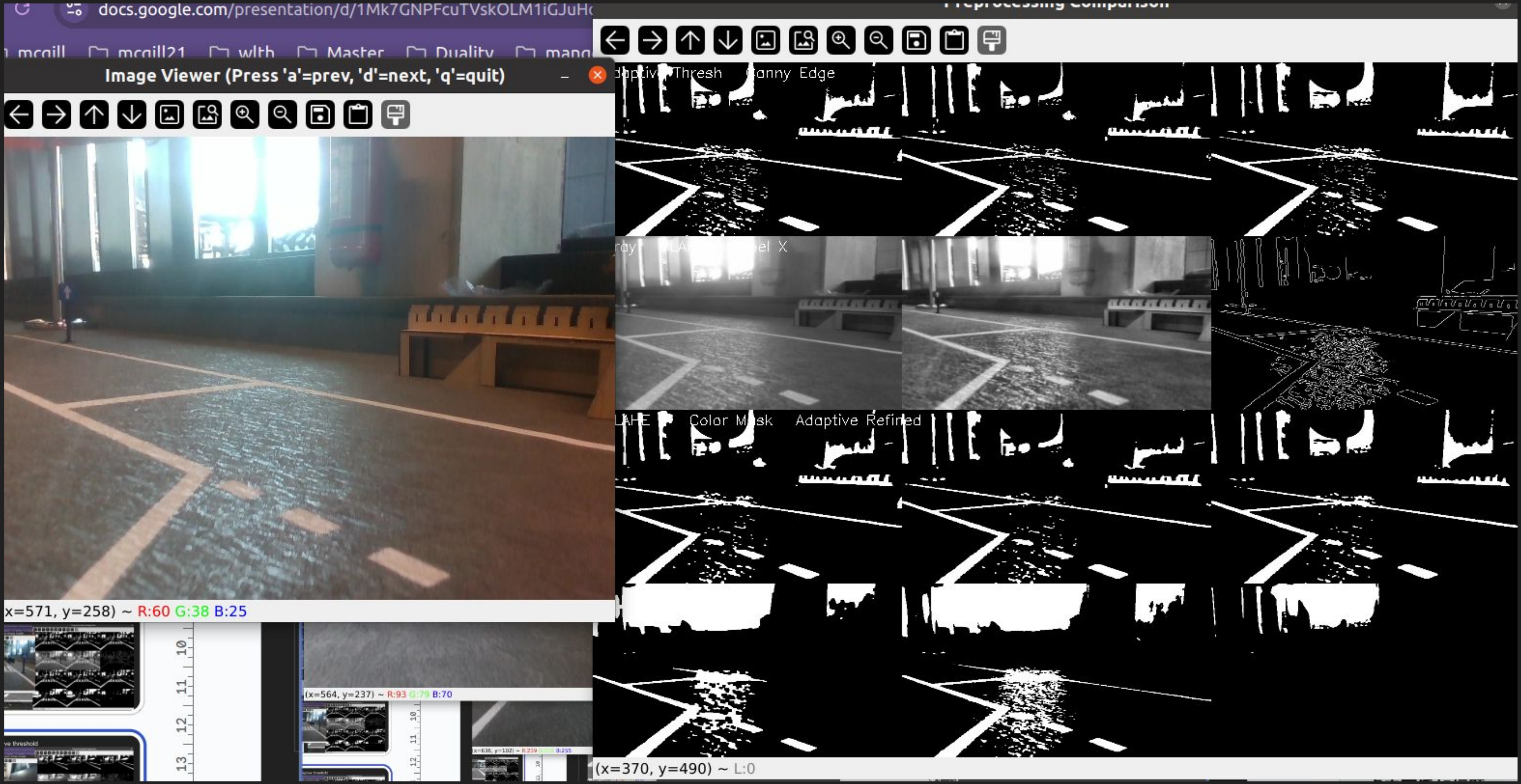
Lane: adaptive threshold



Lane: adaptive threshold



Lane: adaptive threshold



Lane: adaptive threshold reasoning

ChatGPT o3-mini-high

- **Local Calculation:** Instead of using one global threshold, it computes a threshold for small regions of the image.
- **Two Methods:** It can use the mean of the neighborhood (ADAPTIVE_THRESH_MEAN_C) or a weighted sum using a Gaussian window (ADAPTIVE_THRESH_GAUSSIAN_C) to determine the threshold.
- **Constant Subtraction:** A constant value (C) is subtracted from the computed local threshold to fine-tune the result.
- **Binary Output:** Each pixel is set to a maximum value (e.g., 255) if its value is above the local threshold, otherwise it's set to 0.
- **Improved Robustness:** This approach is useful for images with uneven lighting, as it adapts to local variations in brightness.

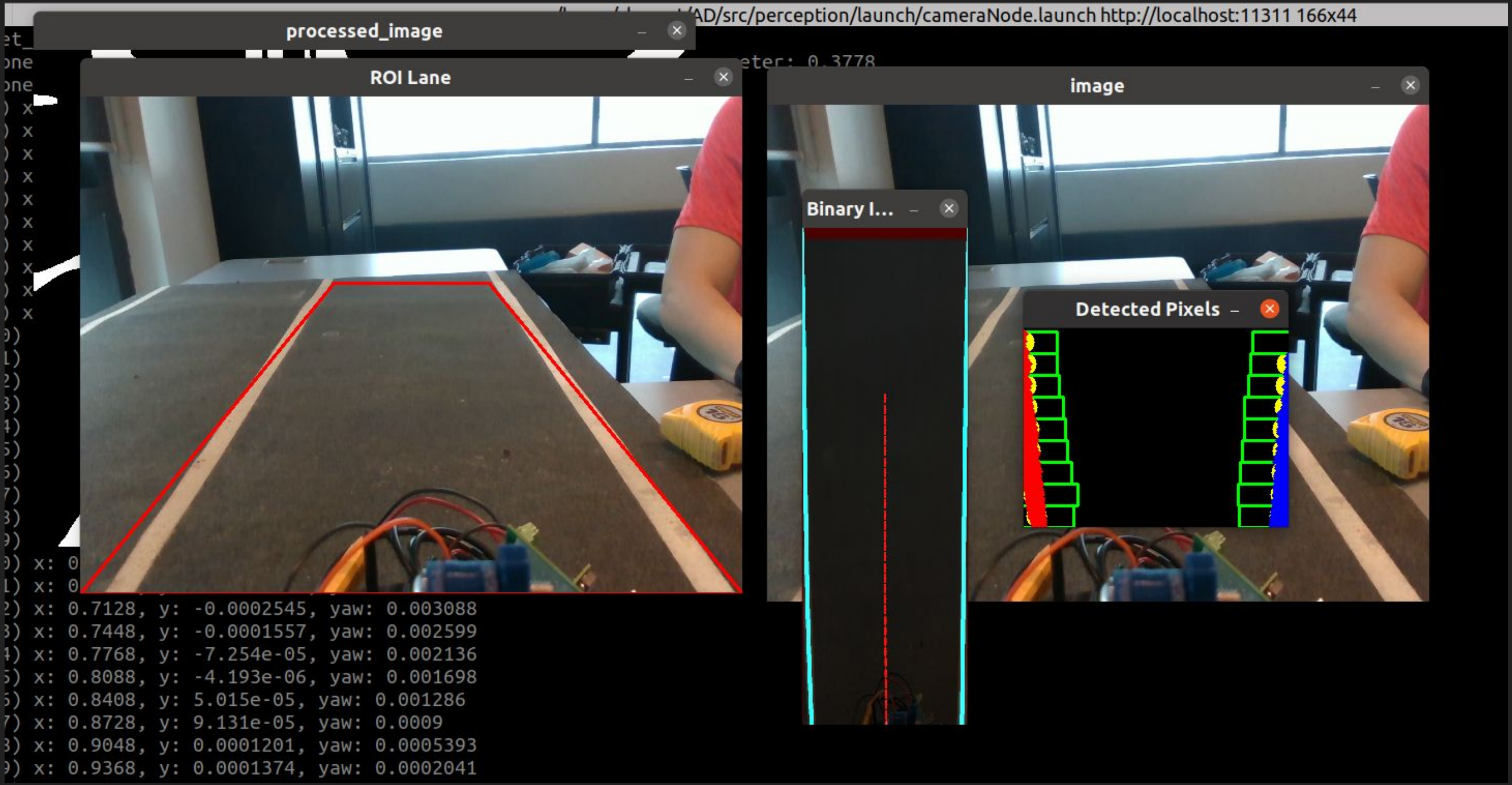
Lane: pixel to meter

estimated	measured	vehicle_x	pixel
	0.35	4.03685	
	0.36	4.02685	
	0.37	4.01685	
	0.38	4.00685	
	0.39	3.99685	
	0.4	3.98685	
	0.41	3.97685	
	0.42	3.96685	
0.06506	0.43	3.95685	15
0.08674	0.44	3.94685	20
0.09758	0.45	3.93685	22.5
0.1084	0.46	3.92685	25
0.1193	0.47	3.91685	27.5
0.141	0.48	3.90685	32.5
0.1518	0.49	3.89685	35
0.1626	0.5	3.88685	37.5
0.1843	0.51	3.87685	42.5
0.2169	0.54	3.84685	50
0.2711	0.57	3.81685	62.5
0.3144	0.6	3.78685	72.5
0.347	0.63	3.75685	80
0.3903	0.66	3.72685	90
0.4337	0.69	3.69685	100

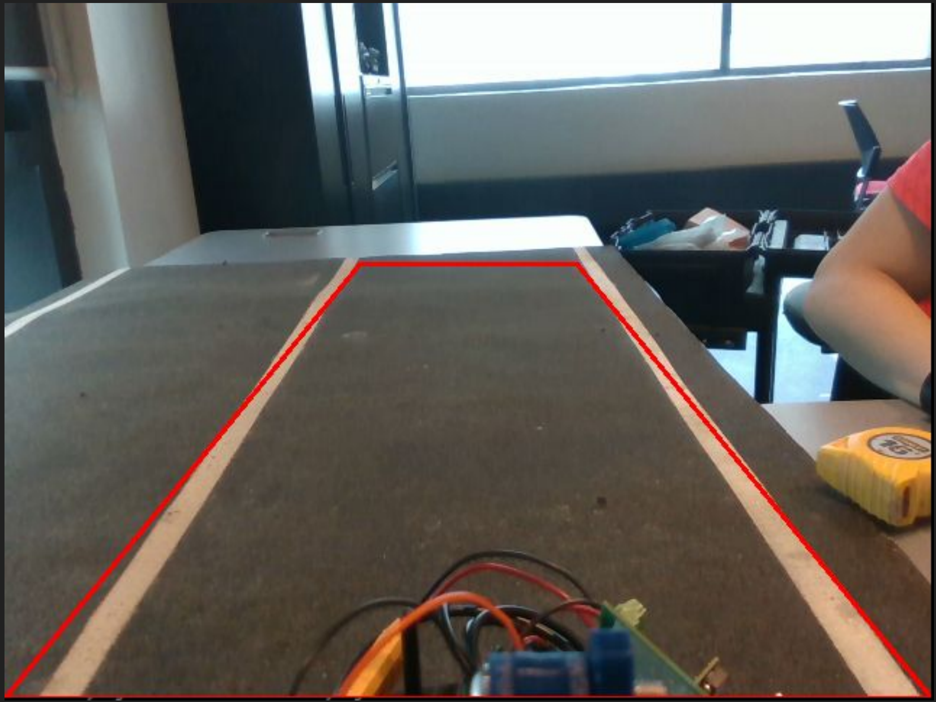
Lane: simulator test

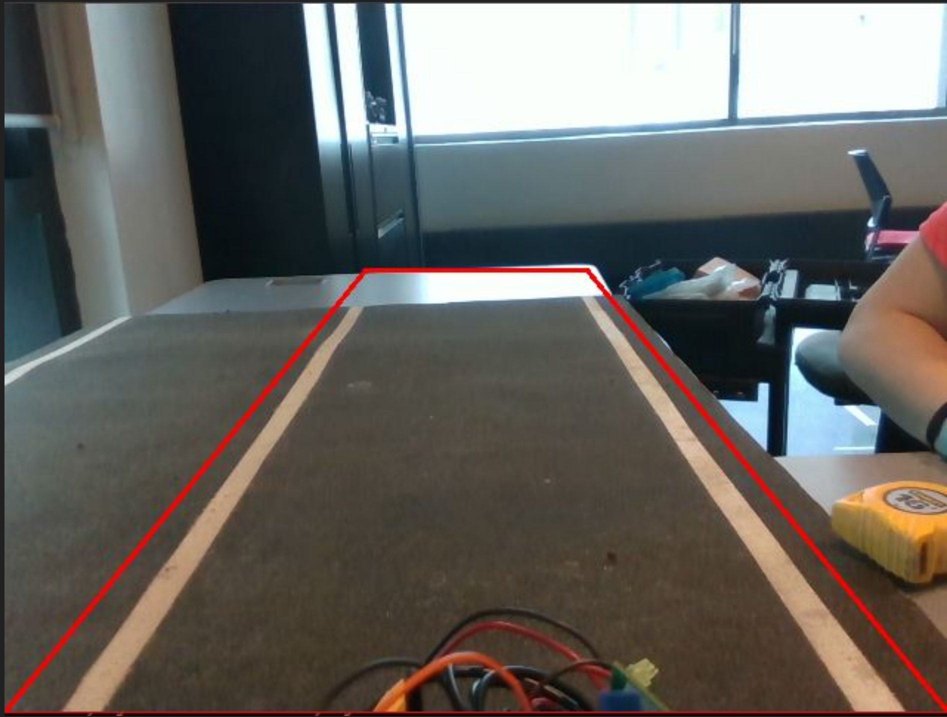
show video

Lane: real world



Lane: real world

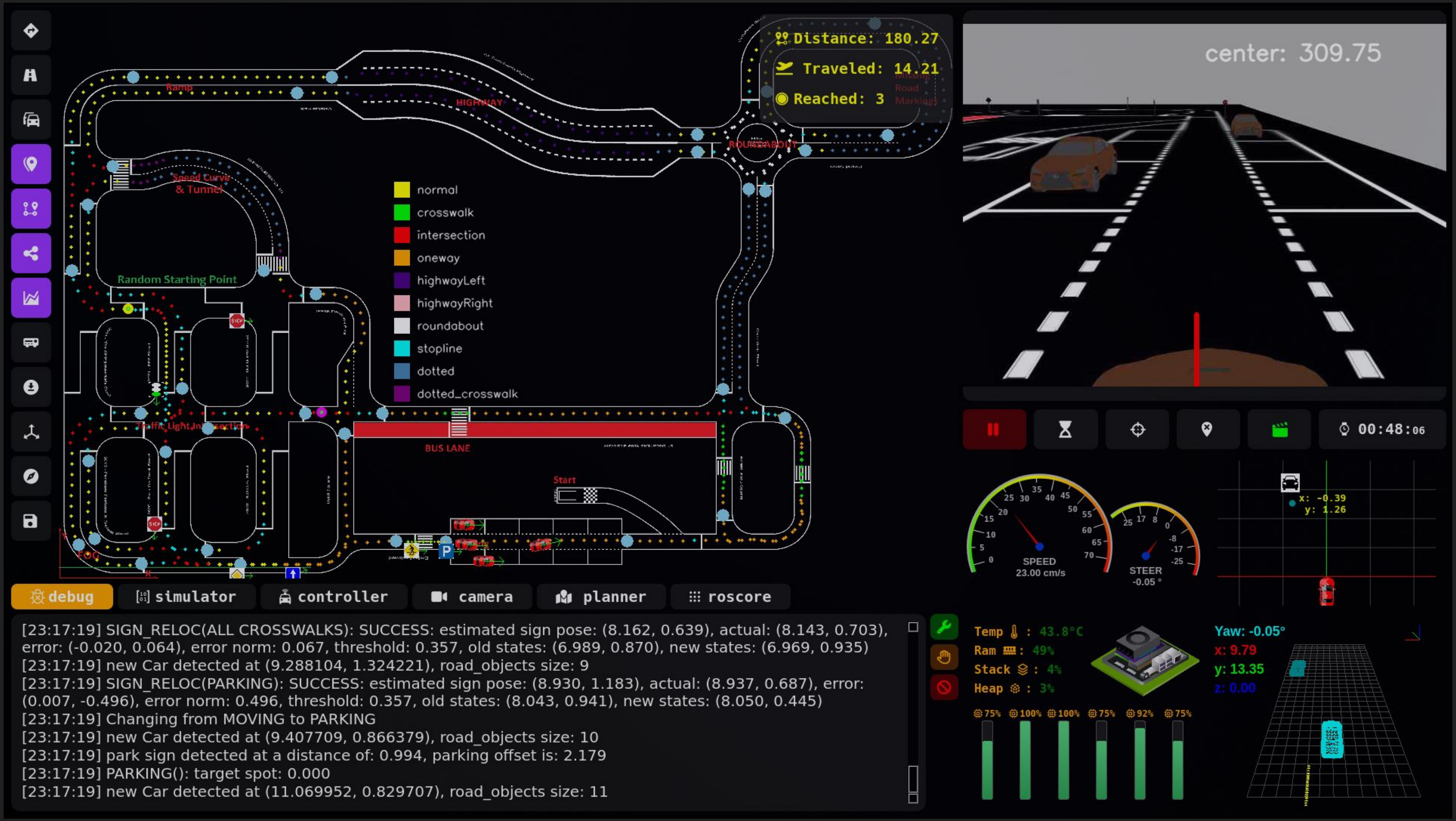




Controller optimizations

- stopline precomputation
- vehicle speed estimation
- overtaking: multi-car scenario

GUI


Distance: 180.27
Traveled: 14.21
Reached: 3
center: 309.75
00:48:06
normal
crosswalk
intersection
oneway
highwayLeft
highwayRight
roundabout
stopline
dotted
dotted_crosswalk
Ramp
HIGHWAY
ROUNDABOUT
Speed Curve & Tunnel
Random Starting Point
BUS LANE
Start
SPEED 23.00 cm/s
STEER -0.05 °
x: -0.39
y: 1.26
debug
simulator
controller
camera
planner
roscore
[23:17:19] SIGN_RELOC(ALL CROSSWALKS): SUCCESS: estimated sign pose: (8.162, 0.639), actual: (8.143, 0.703), error: (-0.020, 0.064), error norm: 0.067, threshold: 0.357, old states: (6.989, 0.870), new states: (6.969, 0.935)
[23:17:19] new Car detected at (9.288104, 1.324221), road_objects size: 9
[23:17:19] SIGN_RELOC(PARKING): SUCCESS: estimated sign pose: (8.930, 1.183), actual: (8.937, 0.687), error: (0.007, -0.496), error norm: 0.496, threshold: 0.357, old states: (8.043, 0.941), new states: (8.050, 0.445)
[23:17:19] Changing from MOVING to PARKING
[23:17:19] new Car detected at (9.407709, 0.866379), road_objects size: 10
[23:17:19] park sign detected at a distance of: 0.994, parking offset is: 2.179
[23:17:19] PARKING(): target spot: 0.000
[23:17:19] new Car detected at (11.069952, 0.829707), road_objects size: 11
Temp : 43.8°C
Ram : 49%
Stack : 4%
Heap : 3%
Yaw: -0.05°
x: 9.79
y: 13.35
z: 0.00

OBJECT DETECTION

- new annotation tool
- new pipeline: label generation + correction
- core test set
- evaluation metrics
- core training set for transparency

OBJECT DETECTION: new pipeline: label generation + correction

live demo

OBJECT DETECTION: new annotation tool

live demo

OBJECT DETECTION: CORE TEST SET

show

OBJECT DETECTION: CORE TRAINING SET FOR TRANSPARENCY

show